Homework 02

REAL VARIABLES I

Fall 2026

Due: 2026-09-06 19:00
Topics: Lebesgue outer measure, outer measures generated by costs, interval coverings, and outer Jordan content

Instructions

Submission is through Gradescope, accessible through Blackboard.
Grading rubric: Every problem/subproblem is graded according to this rubric (no intermediate scores):

- **0%**: No submission or no meaningful content.
- **30%**: Attempt at solution, outline of idea.
- **60%**: Serious attempt at solution. Solution fails to address important points.
- **80%**: Solutions with fixable mistakes or incompleteness.
- **100%**: Complete and correct solution.

You may receive up to a 10% penalty for very unclear or messy exposition.
Please start every solution on a separate page. Clearly mark the pages associated to every part. You may submit either computer typeset solutions or handwritten solutions. Handwritten solutions must be *very clearly* legible.
You are encouraged to work in groups. You may submit the entire homework assignment as a group (up to 4 people). If you only did some exercises as a group, instead of a solution you may clearly indicate whose submission I should look at for the solution to the problem.
Indicate any external help/notes/resources you used. You may not use generative AI in any way to help you with the HW.

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1. (Images of functions with bounded derivative)

(In class.)

1.a

weight: 1.0 Let $f\colon \mathbb{R} \to \mathbb{R}^2$ be differentiable and suppose that its derivative is bounded. Prove that the image $f(\mathbb{R})$ has two-dimensional Lebesgue outer measure zero.

> **Free hint.** First prove that $\mathcal{L}_2^*(f([-N, N])) = 0$ for every $N \in \mathbb{N}$. Then use σ-additivity of outer measures

2. (Equivalent interval definitions of Lebesgue outer measure)

2.a

weight: 1.0 A *dyadic interval* is a half-open interval of the form

$$[k2^{-m}, (k+1)2^{-m}), \qquad k, m \in \mathbb{Z}.$$

On each of the following three collections, assign every interval its length:

1. bounded open intervals;
2. bounded closed intervals;
3. dyadic intervals.

Prove that all three associated generated outer measures are the same. In particular, each one equals the Lebesgue outer measure $\mathcal{L}_1^*$.

3. (Finite interval covers and outer Jordan content)

3.a

weight: 0.3 Prove that $\mathcal{L}_1^*([0,1] \setminus \mathbb{Q}) = 1$

The purpose of this exercise is to show that covering by a finite number of intervals will not suffice in the definition of Lebesgue outer measure.

3.b

weight: 0.7 The *outer Jordan content* of a set $E \subseteq \mathbb{R}$ is defined by

$$J(E) \coloneqq \inf \left\{ \sum_{I \in \mathcal{J}} \ell(I) : \mathcal{J} \text{ is a } \textbf{finite} \text{ collection of bounded closed intervals and } E \subseteq \bigcup_{I \in \mathcal{J}} I \right\}.$$

As usual, $\ell([a,b]) = b - a$.

You do not need to provide a detailed proof of the fact that if $E = [c,d]$ is a bounded closed interval and $[c,d] \subseteq \bigcup_{n=1}^{N} [a_n, b_n]$ for some $N \in \mathbb{N}$ then $\sum_{n=1}^{\infty} (b_n - a_n) \geq d - c$, but you should recall/provide a reference of a fact we proved in class that uses the argument that can be directly adapted to prove this claim.

Prove that $J(E) = J(\bar{E})$. Then exhibit a countable set $E \subseteq \mathbb{R}$ such that $J(E) = 1$, whereas $\mathcal{L}_1^*(E) = 0$.